Name: Beatriz Aimée Teixeira F. Braga

The marked column is the final state of the step. The simulation was stopped when a cycle was encountered.

Time: 0ns
Step: 0

| | | $P_1$ | $P_2$ | $P_3$ | maj |
|---|---|---|---|---|---|
| Init: | H (X,X) | (X,0) | - | - | (0,0) |
| | Q (X,0) | - | - | - | (0,0) |
| | D (X,1) | - | - | - | (1,1) |

nothing left to do

| $P_1$ | $P_2$ | $P_3$ | maj |
|---|---|---|---|
| - | - | - | (0,0) |
| - | - | - | (0,0) |
| - | - | - | (1,1) |

- Wait 10ns

Time: 10ns
Step: 1

| | $P_1$ | $P_2$ | $P_3$ | maj |
|---|---|---|---|---|
| H | (0,1) | - | - | (1,1) |
| Q | - | - | - | (0,0) |
| D | - | - | - | (1,1) |

Flags @H

| $P_1$ | $P_2$ | $P_3$ | maj |
|---|---|---|---|
| - | - | - | (1,1) |
| - | (0,1) | - | (1,1) |
| - | - | - | (1,1) |

nothing left to do

| $P_1$ | $P_2$ | $P_3$ | maj |
|---|---|---|---|
| - | - | - | (1,1) |
| - | - | - | (1,1) |
| - | - | - | (1,1) |

- wait 10ns

Time: 20ns
Step: 2

| | $P_1$ | $P_2$ | $P_3$ | maj |
|---|---|---|---|---|
| H | (1,0) | - | - | (0,0) |
| Q | - | - | - | (1,1) |
| D | - | - | - | (1,1) |

nothing else to do

| $P_1$ | $P_2$ | $P_3$ | maj |
|---|---|---|---|
| ← | - | - | (0,0) |
| - | - | - | (1,1) |
| - | - | - | (1,1) |

- +5s, 15ns after Q was trigged (in 10ns)
- wait 10ns to raise H.

Time: 25ns
Step: 3

| | $P_1$ | $P_2$ | $P_3$ | maj |
|---|---|---|---|---|
| H | - | - | - | (0,0) |
| Q | - | - | - | (1,1) |
| D | - | - | (1,0) | (0,0) |

D ← Q+1

nothing to do

| $P_1$ | $P_2$ | $P_3$ | maj |
|---|---|---|---|
| - | - | - | (0,0) |
| - | - | - | (1,1) |
| - | - | - | (0,0) |

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- Time: 30 ns (10ns after last change in H)
- Step: 4

Flags @H (Q⇐D)

| | P1 | P2 | P3 | maj | P1 | P2 | P3 | maj |
|---|---|---|---|---|---|---|---|---|
| H | (0,1) | – | – | (1,1) | – | – | – | (1,1) |
| Q | – | – | – | (1,1) | – | (1,0) | – | (0,0) |
| D | – | – | – | (0,0) | – | – | – | (0,0) |

- wait 10ns
- Time: 40ns
- step: 5

nothing else

| | P1 | P2 | P3 | maj | P1 | P2 | P3 | maj |
|---|---|---|---|---|---|---|---|---|
| H | (1,0) | – | – | (0,0) | – | – | – | (0,0) |
| Q | – | – | – | (0,0) | – | – | – | (0,0) |
| D | – | – | – | (0,0) | – | – | – | (0,0) |

+5ns, 15ns after Q⇐1.

- Time: 45ns
- step: 6

D⇐Q+1 = 0+1

| | P1 | P2 | P3 | maj | P1 | P2 | P3 | maj |
|---|---|---|---|---|---|---|---|---|
| H | – | – | – | (0,0) | – | – | – | (0,0) |
| Q | – | – | – | (0,0) | – | – | – | (0,0) |
| D | – | – | (0,1) | (1,1) | – | – | – | (1,1) |

+5ns (10ns after last change in H)

- Time: 50ns
- step 7

@H(Q⇐1)

| | P1 | P2 | P3 | maj | P1 | P2 | P3 | maj | P1 | P2 | P3 | maj |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| H | (0,1) | – | – | (1,1) | – | – | – | (1,1) | – | – | – | (1,1) |
| Q | – | – | – | (0,0) | – | (0,1) | – | (1,1) | – | – | – | (1,1) |
| D | – | – | – | (1,1) | – | – | – | (1,1) | – | – | – | (1,1) |

+10ns

- Time: 60ns
- Step 8

| | P1 | P2 | P3 | maj | P1 | P2 | P3 | maj |
|---|---|---|---|---|---|---|---|---|
| H | (1,0) | – | – | (0,0) | – | – | – | (0,0) |
| Q | – | – | – | (1,1) | – | – | – | (1,1) |
| D | – | – | – | (1,1) | – | – | – | (1,1) |

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wait + 5ns (13ns after Q trig)

- time: 65ns
- step: 9

| | $P_1$ | $P_2$ | $P_3$ | maj |
|---|---|---|---|---|
| H | – | – | – | (0,0) |
| Q | – | – | – | (1,1) |
| D | – | – | (1,0) | (0,0) |

| $P_1$ | $P_2$ | $P_3$ | maj |
|---|---|---|---|
| – | – | – | (0,0) |
| – | – | – | (1,1) |
| – | – | – | (0,0) |

wait +5 ns (10 ns after last change in H)

- time: 70ns
- step: 10

@H, Q⇐0

| | $P_1$ | $P_2$ | $P_3$ | maj |
|---|---|---|---|---|
| H | (0,1) | – | – | (1,1) |
| Q | – | – | – | (1,1) |
| D | – | – | – | (0,0) |

| $P_1$ | $P_2$ | $P_3$ | maj |
|---|---|---|---|
| – | – | – | (1,1) |
| – | (1,0) | – | (0,0) |
| – | – | – | (0,0) |

wait 10ns

- time 80ns
- step: 11

| | $P_1$ | $P_2$ | $P_3$ | maj |
|---|---|---|---|---|
| H | (1,0) | – | – | (0,0) |
| Q | – | – | – | (0,0) |
| D | – | – | – | (0,0) |

| $P_1$ | $P_2$ | $P_3$ | maj |
|---|---|---|---|
| – | – | – | (0,0) |
| – | – | – | (0,0) |
| – | – | – | (0,0) |

⇓

cycle found!

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